AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 21 · WORKBOOK

Similarity by Selected Traits

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected distance-comparison record.

Read vectors, compute gaps, repair feature order and limit exact-match claims.

Keep one habit: Keep feature definitions, target, candidate coordinates, scaling, weights and all ties visible.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Repair Bias in Data. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Repair Bias in Data

Missing Data

Choose Informative Questions

W1 · Read the declared vector order

> Position0 and1 have different meanings.

Your evidence record

- Activity reference: feature order energy,group work; each0–5. Original target[2,4], Sketch[1,3], Relay[5,4]; Manhattan distance sums absolute gaps with equal weights. W2 supplies a different target[1,2] and its own candidates. Book examples use a separate[2,5] target and declared pace/illustration features. Zero distance means matching recorded coordinates only.

- For [4, 1] under the worked feature order, identify energy and group-work ratings.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Compute both coordinate gaps

> Use the target and options supplied in this task.

Your evidence record

- Activity reference: feature order energy,group work; each0–5. Original target[2,4], Sketch[1,3], Relay[5,4]; Manhattan distance sums absolute gaps with equal weights. W2 supplies a different target[1,2] and its own candidates. Book examples use a separate[2,5] target and declared pace/illustration features. Zero distance means matching recorded coordinates only.

- From target [1, 2], find **distance**s to A [2, 2] and B [0, 4].

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair inconsistent feature order

> Reconcile the schema before calculating distance.

Your evidence record

- Activity reference: feature order energy,group work; each0–5. Original target[2,4], Sketch[1,3], Relay[5,4]; Manhattan distance sums absolute gaps with equal weights. W2 supplies a different target[1,2] and its own candidates. Book examples use a separate[2,5] target and declared pace/illustration features. Zero distance means matching recorded coordinates only.

- One record uses [group work, energy] while others use [energy, group work]. Why is the **distance** unreliable?

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Limit distance-zero claims

> Unrecorded requirements can still differ.

Your evidence record

- Activity reference: feature order energy,group work; each0–5. Original target[2,4], Sketch[1,3], Relay[5,4]; Manhattan distance sums absolute gaps with equal weights. W2 supplies a different target[1,2] and its own candidates. Book examples use a separate[2,5] target and declared pace/illustration features. Zero distance means matching recorded coordinates only.

- Two activities have **distance** zero. Must they suit every learner equally well?

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Rank independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Use the given target[3,2] and its own options with equal-weight Manhattan distance; preserve all tied distances. Do not reuse the worked target or weight change.

Target is [3, 2]. Options A [1, 2], B [3, 4], C [4, 2]. Calculate **distance**s, select the closest, and identify a tie among the others.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Rank independently

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Make selected traits and their **scale**s visible in every recommendation explanation.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Compute **distance**s for **Distance** between **feature vector**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Distance** between **feature vector**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Represent books as [animal,short,illustrated] with 1=yes. Query=[1,1,0]; A=[1,0,0], B=[1,1,1], C=[0,0,0]. Count mismatches. What changes if illustrations matter twice as much?

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Traits are [animals,adventure]. A=[1,1], B=[1,0], C=[0,0]; query=[1,0].

Use number of differing traits as **distance**.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

Score **similarity** using one point for same topic and one for same difficulty. Target is space, easy. A is space, hard; B is animals, easy. What happens?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 21 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Reference · Original target response space

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Represent books as [animal,short,illustrated] with 1=yes. Query=[1,1,0]; A=[1,0,0], B=[1,1,1], C=[0,0,0]. Count mismatches. What changes if illustrations matter twice as much?

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Target is [3, 2]. Options A [1, 2], B [3, 4], C [4, 2]. Calculate **distance**s, select the closest, and identify a tie among the others.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

> Keep the explanation and the support you used.

Use this when helpful

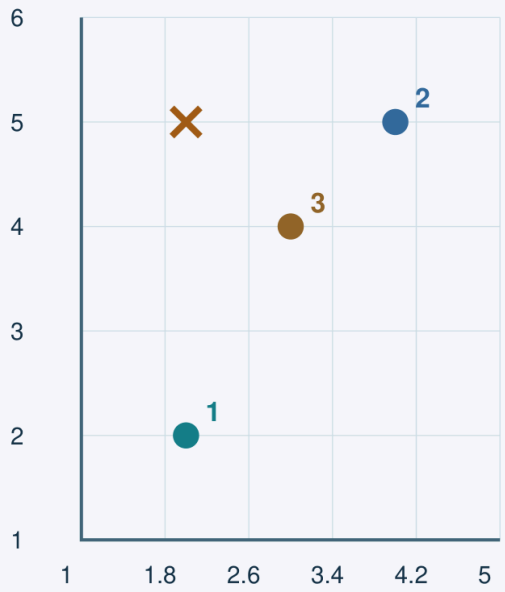
A fictional toy book matcher represents two preferences with numeric features.

Target vector [2,5]. Candidate A=[2,2], B=[4,5], C=[3,4].

Use **Manhattan distance**: add absolute **coordinate** differences; return all smallest-**distance** candidates.

AI ACADEMY / AI-2/21

Toy feature positions



Horizontal: first toy feature; vertical: second toy feature

1: A at (2, 2)

2: B at (4, 5)

3: C at (3, 4)

Cross: Target at (2, 5)

Distance rule: Manhattan distance; horizontal=first feature, vertical=second feature

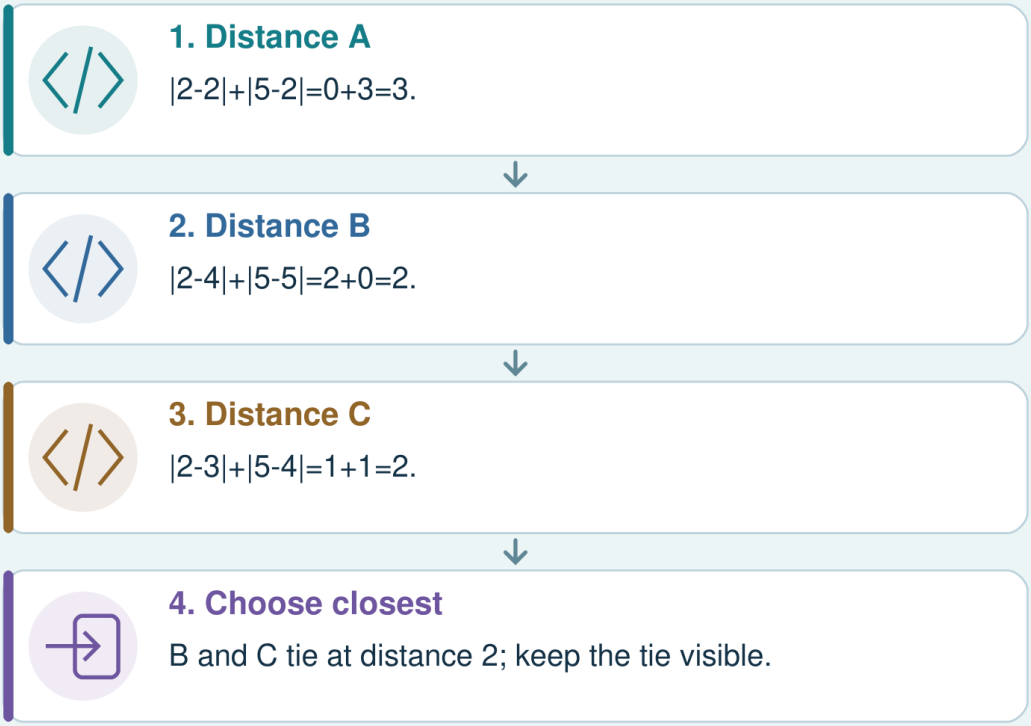
Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/21

Similarity by Selected Traits

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



CHANGE ONE THING

Give the first-coordinate difference weight 2 while the second keeps weight 1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why B ceases to be a closest option without any candidate data changing.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Specify recommender feature meanings, units, scaling and tie policy before explaining nearest matches.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Transparent Book Recommender

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Specify recommender feature meanings, units, scaling and tie policy before explaining nearest matches.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional toy book matcher represents two preferences with numeric features.
- Which labelled step matches this detail? B and C tie at **distance** 2; keep the tie visible.
- 1. **Distance** C
- 2. Choose closest
- 3. **Distance** A
- 4. **Distance** B
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Choose closest B and C tie at **distance** 2; keep the tie visible. Each **coordinate** contributes according to the declared **distance** rule.

Smaller **distance** means closer in this representation, not automatically a better reading experience. If one **coordinate** uses a much larger numerical **scale**, it can dominate the sum even when it is not more important.

Feature units and scaling belong in the method record.

Changed-case question

- Changed-case practice: Give the first-**coordinate** difference weight 2 while the second keeps weight 1. Which conclusion is supported by the supplied facts?
- 1. **Distance**s become A=3, B=4, C=3; A and C tie.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: **Distance**s become A=3, B=4, C=3; A and C tie. The representation's comparison rule changes; the supplied vectors stay the same.

Repair and save your work

Review Repair Bias in Data (ai-2/20). Goal: Identify selection and coverage bias. Short practice: Why must 35% be labelled an illustration rather than the measured grade-wide Yes rate?.

Return to this week and explain what you changed.

Review Missing Data (ai-2/18). Goal: Detect absent and unusable values. Short practice: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?. Return to this week and explain what you changed.

Review Choose Informative Questions (ai-2/15). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Choose the best original question and explain why its more balanced split matters here..

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Similarity by Selected Traits: the case

AI ACADEMY / AI-2/21 · 01 · LOOK AT THE FACTS

Similarity by Selected Traits: the case

The situation

A fictional toy book matcher represents two preferences with numeric features.

Supplied evidence

Target vector [2,5]. Candidate A=[2,2], B=[4,5], C=[3,4].
Use Manhattan distance: add absolute coordinate differences; return all smallest-distance candidates.

By the end, I can calculate a feature distance and state how scaling changes its meaning.

Follow the labelled picture with the matching explanation. Open the vector diagram

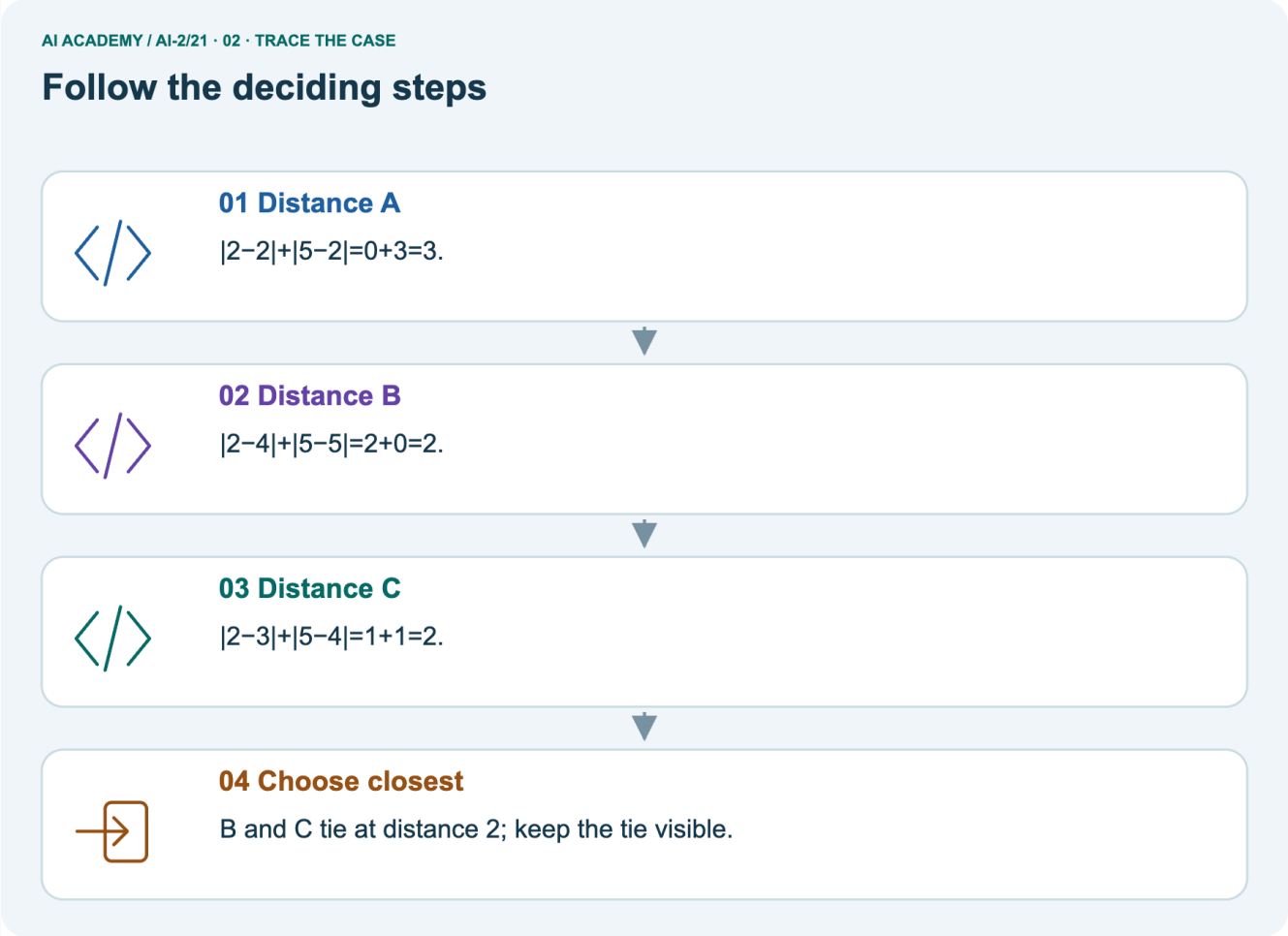
A fictional toy book matcher represents two preferences with numeric features. Supplied facts: Target vector [2,5]. Candidate A=[2,2], B=[4,5], C=[3,4].

Use **Manhattan distance**: add absolute **coordinate** differences; return all smallest-**distance** candidates.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Distance A: |2−2|+|5−2|=0+3=3. → **Distance** B: |2−4|+|5−5|=2+0=2. → **Distance** C: |2−3|+|5−4|=1+1=2. → Choose closest: B and C tie at **distance** 2; keep the tie visible.

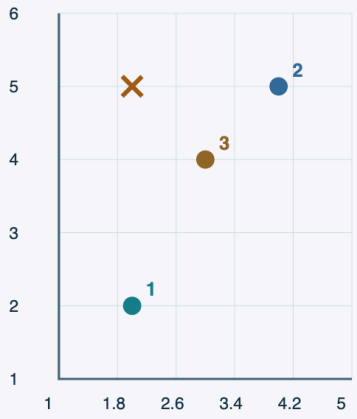
Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Toy feature positions

AI ACADEMY / AI-2/21

Toy feature positions



Horizontal: first toy feature; vertical: second toy feature

1: A at (2, 2)

2: B at (4, 5)

3: C at (3, 4)

Cross: Target at (2, 5)

Distance rule: Manhattan distance; horizontal=first feature, vertical=second feature

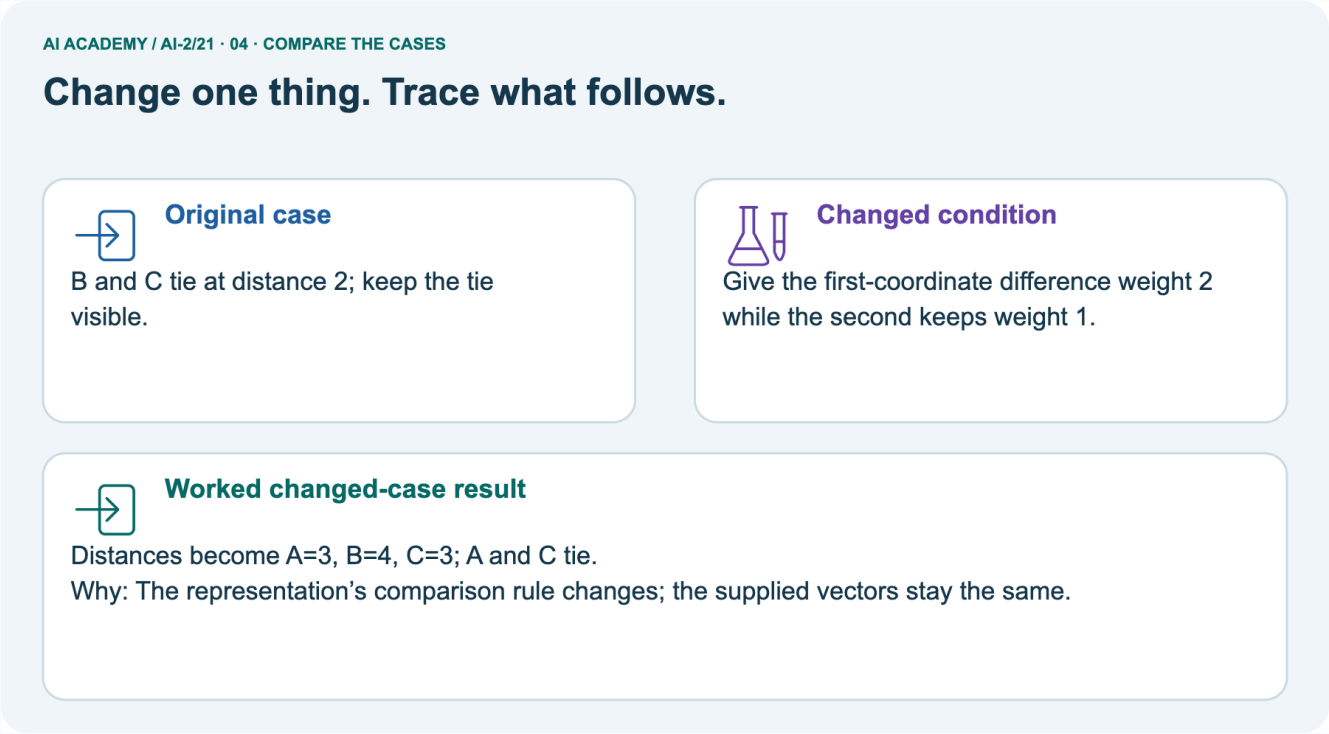
Follow the labelled picture with the matching explanation. Open the vector diagram

Toy feature positions. A: (2, 2). B: (4, 5). C: (3, 4). Query Target: (2, 5). Axes: first toy feature; second toy feature. **Distance**: **Manhattan distance**; horizontal=first feature, vertical=second feature.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

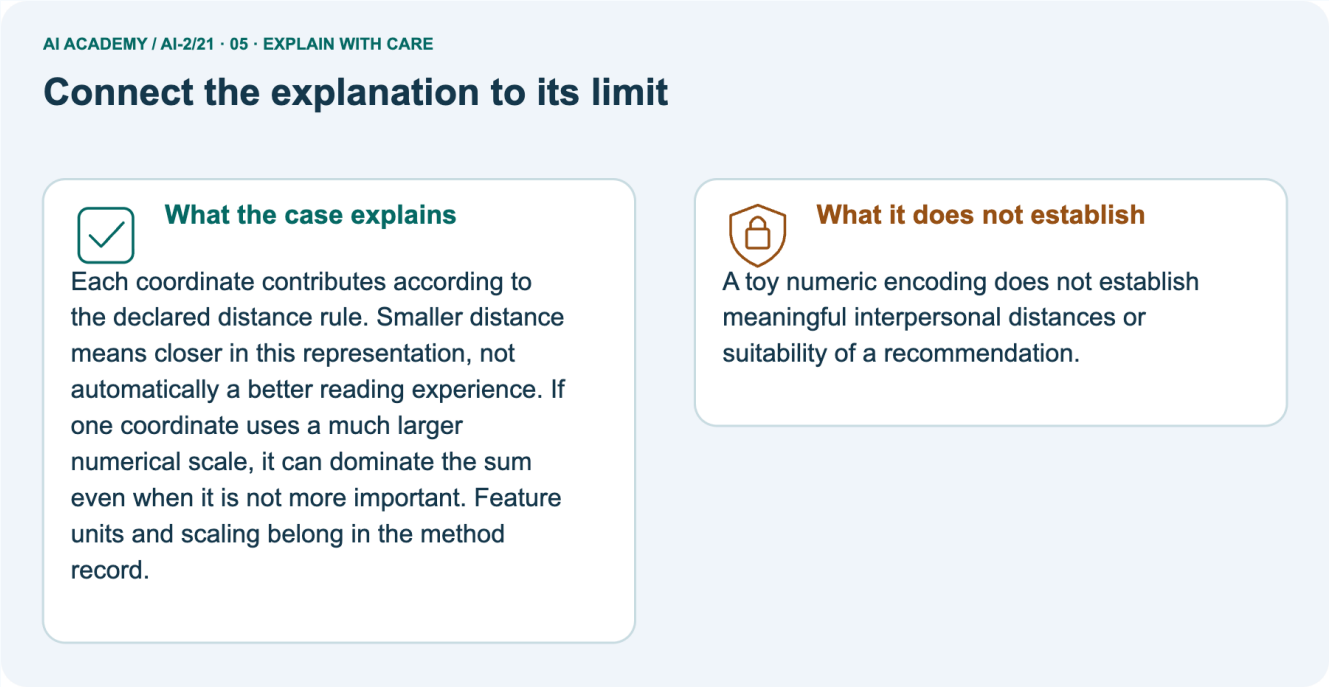
Original case: B and C tie at **distance** 2; keep the tie visible. Changed condition: Give the first-**coordinate** difference weight 2 while the second keeps weight 1. Expected result: **Distance**s become A=3, B=4, C=3; A and C tie.

Reason: The representation's comparison rule changes; the supplied vectors stay the same.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Each **coordinate** contributes according to the declared **distance** rule. Smaller **distance** means closer in this representation, not automatically a better reading experience.

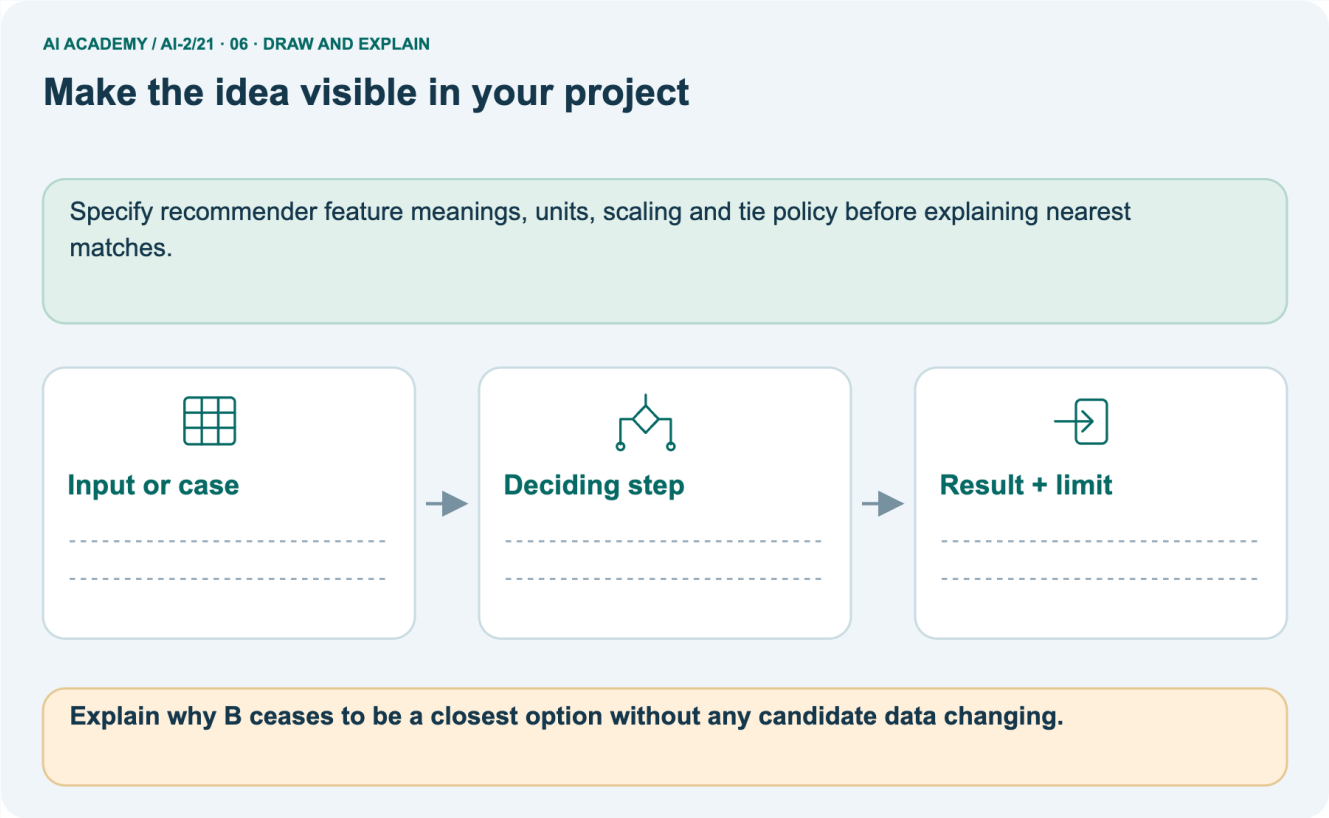
If one **coordinate** uses a much larger numerical **scale**, it can dominate the sum even when it is not more important. Feature units and scaling belong in the method record.

Limit: A toy numeric encoding does not establish meaningful interpersonal **distance**s or suitability of a recommendation.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Specify recommender feature meanings, units, scaling and tie policy before explaining nearest matches. Draw your **input**, deciding step, result and limit.

Supported question: Explain why B ceases to be a closest option without any candidate data changing.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Similarity by Selected Traits: the case

02. Follow the deciding steps

03. Toy feature positions

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.